Fifty Nifty Variations of Two-Transistor Circuits: A tribute to the versatility of MOSFETs

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***Abstract*—We present a compendium of two-MOS-transistor circuits, spanning the range from simple standard configurations to ingenious arrangements. Using these building blocks, circuit designers can assemble a vast array of complex analog functions. This (incomplete) collection shall serve as a reference and inspiration to junior circuit designers and hopefully contains at least one unexpected example for the professional engineer.**

***Index Terms*—Basic circuits, body-driven, CMOS, MOSFET.**

I. Introduction

ANALOG circuit design is wonderfully creative. The metal-oxide-semiconductor field-effect transistor (MOSFET) is an exceptionally versatile device, operating as a switch, a current source, a resistor, a diode, and a capacitor, depending on bias conditions. For fun and to demonstrate the sheer infinite possibilities in circuit design using MOSFET, we present a collection of simple (and sophisticated) circuits, which employ *two* transistors (not counting fixed bias and supply voltages and fixed bias currents). Often, circuit designers construct complex circuits from these basic building blocks.

This compendium is a tribute to all the ingenious minds out there and the circuit design giants on whose shoulders we are standing today! This sample of practical two-transistor circuits, to the best of the authors' knowledge, contains beneficial and often used configurations. A few circuits are of a more curious and academic nature, they might lack power-supply rejection or show other deficiencies, and some circuits use the body connection as active terminals (Figs. 2, 16, 27, 29, 46, 49 and 50), which might not be feasible in some complementary metal-oxide-semiconductor (CMOS) technologies. Generally, one has to be aware of the body effect and its impact.

Many more two-transistor circuits are yet to be discovered. An exhaustive search of graphs using one or two voltage-controlled current sources (which are well approximated by MOSFET) resulted in 150 potentially useful circuits [1], where one of them was identified as a valuable new amplifier configuration [2]. By pushing this idea further, a study identified 582 possible circuit topologies using two transistors—repeating this exercise using *three* transistors, a whopping 56 280 elementary configurations have been found [3].

To keep our overview reasonable, we do not include complementary circuits which can be constructed by swapping n-MOSFET (NMOS) for p-MOSFET (PMOS) devices (or vice versa). This can be applied to Figs. 3 to 13, 15 to 18, 20 to 42 and 44 to 48.

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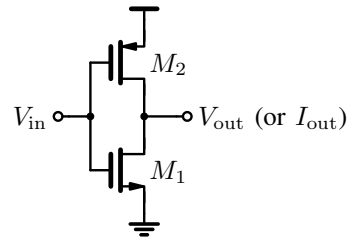

Fig. 1. The ubiquitous digital inverter. The input voltage V_{in} switches one of both transistors on, and the other is off [4].

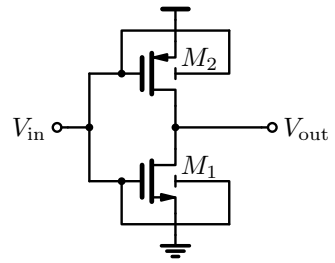

Fig. 2. The DTMOS inverter achieves an improved current drive at low leakage current. It needs to be operated at low supply voltages to avoid a forward bias of the well diodes [5].

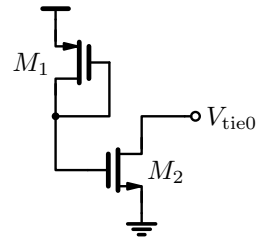

Fig. 3. An ESD-safe tie-zero for unused CMOS logic inputs (no MOSFET gate is tied directly to a supply rail). The tie-one can be constructed accordingly.

II. Logic Circuits

First, we present simple circuits which operate on logical inputs, like the inverter (Fig. 1) and an improved low-voltage version that is exploiting the body terminals in a dynamic threshold voltage MOSFET (DTMOS) inverter (Fig. 2). Terminating an unused input of a logic gate to a logical one or zero, an electrostatic discharge (ESD)-safe tie-zero or tie-one, respectively, should be utilized like the one shown in Fig. 3.

Using the unusual arrangement in Fig. 4, we can implement an XNOR logic function, while the implementations of NAND and NOR, as shown in Figs. 5 and 6, respectively, are the fundamental building blocks of the digital universe.

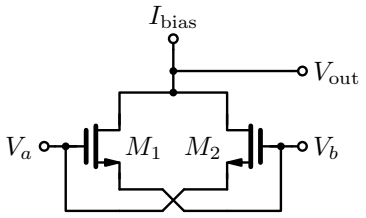

Fig. 4. Using a current source I_{bias} with finite output impedance to bias this structure, this circuit implements an XNOR logic function ($V_{\text{out}} = \overline{V_a \oplus V_b}$). The logic inputs V_a and V_b must be driven by low-ohmic logic levels between V_{DD} and V_{SS} [6].

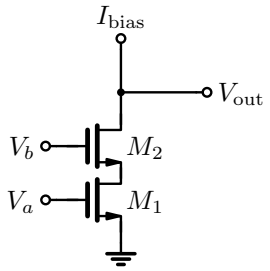

Fig. 5. This series connection of two MOSFETs realizes a logical NAND function ($V_{\text{out}} = \overline{V_a \wedge V_b}$).

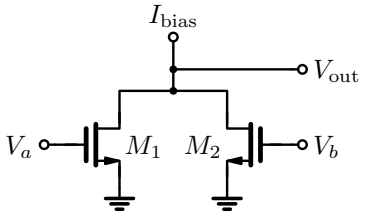

Fig. 6. This circuit complements the logic gates implemented in Figs. 4 and 5 and realizes a NOR function ($V_{\text{out}} = \overline{V_a \vee V_b}$).

III. Basic Circuits

This section presents the basic circuits constituting the fundamental building blocks of the analog and mixed-signal world, like the current mirror shown in Fig. 7. The differential pair (Fig. 8) is alternatively constructed with or without a tail current source, often called a pseudo-differential pair. The source follower in Fig. 9 is one of the elementary circuits useful in many situations, just like the common-source amplifier with active load (Fig. 10). Adding a cascode to a circuit is a powerful technique. For example, a common-gate combined with a common-source stage is depicted in Fig. 11, or the cascade of two common-gate stages is shown in Fig. 12. Of course, the essential common-gate stage qualifies as a two-transistor circuit as well (Fig. 13).

The transmission gate (Fig. 14) is an essential building block, allowing switching voltages and currents (a feat making the MOSFET such a valuable device). With a straightforward extension, a multiplexer can be constructed (Fig. 15). However, in most practical implementations, a set of transmission gates will be applied for this purpose.

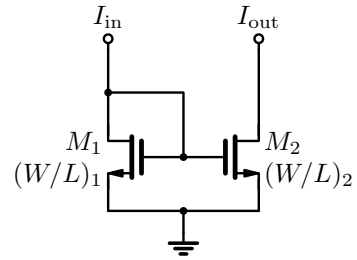

Fig. 7. The basic current mirror, simultaneously copying and sizing of $I_{\text{out}} = (W/L)_2/(W/L)_1 \cdot I_{\text{in}}$ according to the dimensions of M_1 and M_2 [7].

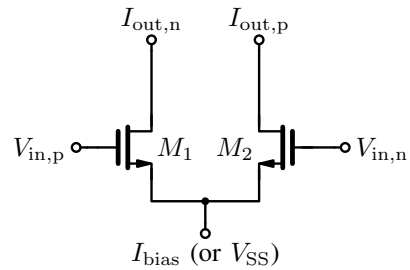

Fig. 8. The ubiquitous differential pair, like the current mirror in Fig. 7, is a fundamental building block in integrated circuits [8]. The pseudo-differential variant spares the tail current source's headroom in exchange for reduced common-mode rejection, but with the benefit of class-AB action.

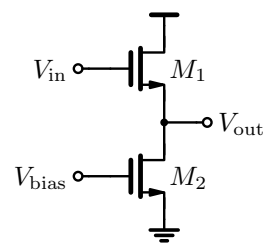

Fig. 9. The source follower (or common-drain stage), utilizing M_2 as a current source to bias M_1.

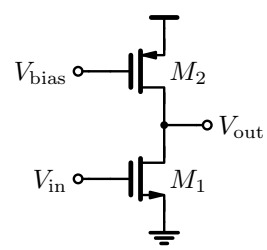

Fig. 10. The common-source amplifier with active load.

IV. Improved Basic Circuits

It has been realized that the digital inverter shown in Fig. 1 can also be employed as an excellent low-voltage amplifier (V_{out}) or transconductance stage (I_{out}) when both transistors are biased in saturation. Shorting V_{in} and V_{out}, a replica bias is readily available [9].

Several low-voltage circuits can be implemented using the body terminal of a MOSFET as an additional control input, like the low-voltage current mirror depicted in Fig. 16. Some arrangements allow for precise voltage gain and high-speed

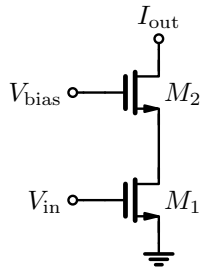

Fig. 11. The cascoded common-source stage boosting the output impedance of M_1 considerably to $r_{\rm out} \approx g_{m2}/(g_{ds1} \cdot g_{ds2})$.

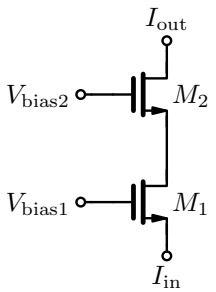

Fig. 12. The cascoded common-gate stage. Note that $I_{\rm out} \approx I_{\rm in}$, but the impedance level changes drastically, creating gain or a high output impedance at the output node.

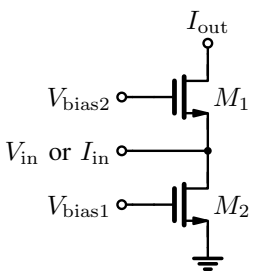

Fig. 13. The common-gate stage, employing M_2 as a current source to bias transistor M_1.

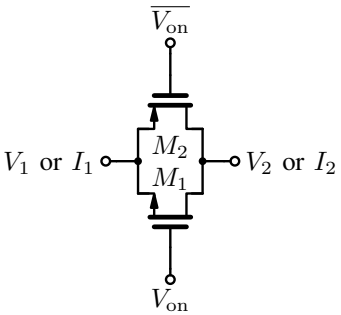

Fig. 14. The transmission gate switches between V_1/I_1 and V_2/I_2 for both voltage and current (and it works rail to rail, too).

operation, as demonstrated by the circuits shown in Figs. 17 and 18.

Figure 19 presents an enhanced version of the source follower, also known as class-B (push/pull) amplifier. Degeneration is a primary method to improve matching, noise figure, or output resistance. Different MOSFET-only implementations

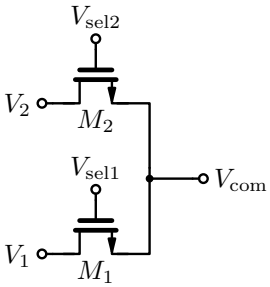

Fig. 15. The 2-to-1 multiplexer, connecting either V_1 or V_2 to $V_{\rm com}$. Depending on $V_{\rm sel1}$ and $V_{\rm sel2}$, the MOSFETs are alternately switched on or off.

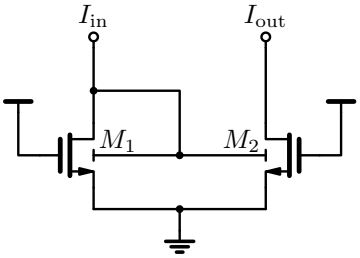

Fig. 16. This circuit is an improved version of Fig. 7 in that it allows a low voltage operation of the current mirror, requiring a voltage headroom substantially less than $V_{\rm GS1}$ [12].

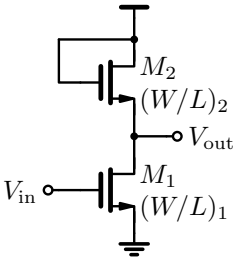

Fig. 17. The common-source amplifier with diode load is sometimes called a wide-band amplifier due to its potentially high-speed operation. Here, the gain is set precisely at $A_v = V_{\rm out}/V_{\rm in} = -\sqrt{(W/L)_1/(W/L)_2}$, only depending on transistor sizing (and neglecting the body effect).

are feasible (Figs. 20 and 21), with the implementation in Fig. 21 allowing to tune the performance during operation by varying $V_{\rm bias}$.

Combining a common-source stage with a common-gate topology [10], as shown in Fig. 22, results in a differential-output transconductance stage with a single-ended low-ohmic input. It can thus be used under impedance-matched conditions, also providing some noise and linearity cancellation under ideal bias [11].

In yet another twist, a source-follower can be combined with a common-gate stage, resulting in a useful amplifier configuration (see Fig. 23), which has been found by systematically generating graphs consisting of two transconductance stages [2].

V. Biasing Circuits

The generation of a bias voltage is a task often encountered in analog circuit design. When the requirements on stability

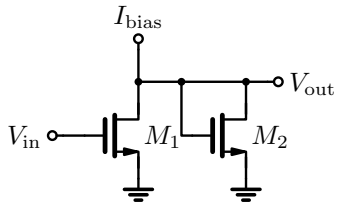

Fig. 18. The folded version of Fig. 17, having the advantage of a removed body effect in M_2 and a ground-referred output node [13].

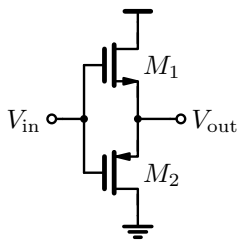

Fig. 19. The class-B push-pull follower can be considered an enhanced version of the simple source follower shown in Fig. 9. Lacking a class-A bias component, this structure is subjected to cross-over distortion.

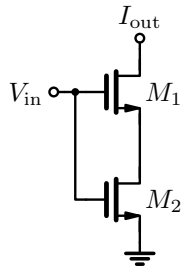

Fig. 20. The MOSFET-R degenerated common-source stage. By sizing M_2 appropriately, the degeneration can be adapted [13]. This arrangement using two transistors can also increase the length of a (compound) device, for example, in current mirrors, as otherwise, MOSFET with different L will not match well.

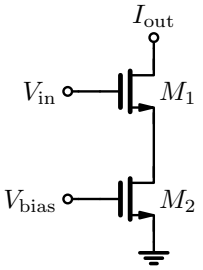

Fig. 21. A variation of the implementation shown in Fig. 20, where the degeneration of M_1 can be adapted by tuning V_{bias}.

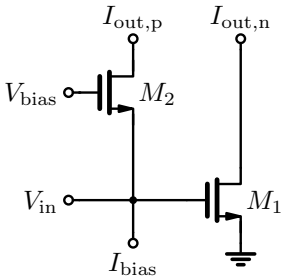

Fig. 22. A common-gate–common-source topology offering impedance-matched single-ended input and differential output while simultaneously canceling noise and distortion [11].

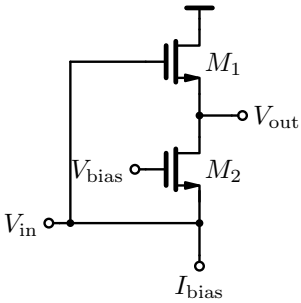

Fig. 23. This low-noise amplifier was discovered by using an exhaustive search of potential two-transistor wide-band amplifiers. For practical implementation, M_1 requires an ac coupling (and proper biasing) in its gate connection to keep M_2 in saturation [2].

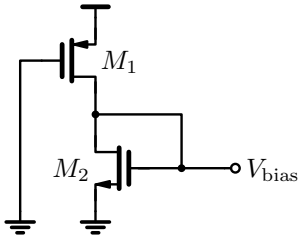

Fig. 24. A (simple) bias voltage generator using the current source M_1 to bias M_2 so that $V_{\text{bias}} = V_{\text{GS2}}$.

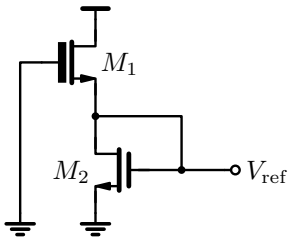

Fig. 25. A constant bias voltage generator (M_1 and M_2 must have different threshold voltages $V_{\text{th1}} \neq V_{\text{th2}}$) [14].

are moderate, a simple configuration, as shown in Fig. 24, might be sufficient. A surprisingly stable voltage reference can be constructed from two MOSFETs with different threshold voltages, as demonstrated in Fig. 25. Occasionally, a bias voltage with a well-defined proportionality to temperature is needed, for example, in temperature sensor circuits. This effect can be achieved by a proportional-to-absolute temperature (PTAT) voltage generator like the one depicted in Fig. 26. Sometimes, circuits are based on obscure second-order effects, like charge trapping in the Si–SiO_2 interface traps, used in the clocked circuit shown in Fig. 27, which can create pA-currents in an area-efficient manner.

VI. Diverse Circuit Elements

The various operating modes of a MOSFET can be utilized differently and lead to practical circuit elements. The cross-coupled differential pair (see Figs. 28 and 29) synthesizes

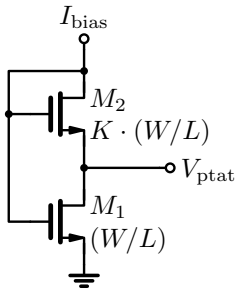

Fig. 26. A PTAT voltage generator, if M_1 and M_2 are kept in subthreshold operation [15].

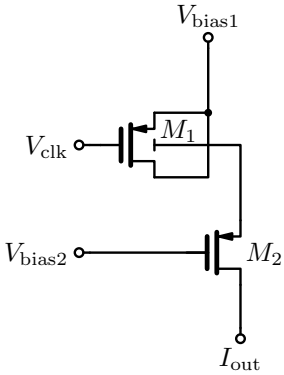

Fig. 27. This pA current source is based on the periodic filling and flushing of Si–SiO_2 interface traps by alternating M_1 between accumulation and inversion (by proper choice of V_{bias1}, V_{bias2}, and the switching levels of V_{clk}). It can operate with reasonably high clock frequencies and still create tiny currents [16].

a negative resistance, which can cancel losses, for example, in oscillators or Q-enhanced LC filters. Since the effective capacitance between the gate terminal and the source/drain connection of a MOSFET is a function of biasing conditions, the arrangement demonstrated in Fig. 30 can be employed as a varactor.

Using local feedback to create a MOSFET-"diode," two anti-parallel diodes can replace conventional *pn*-diodes, for example, in voltage limiters (Fig. 31). The gate oxide of a MOSFET usually offers the highest capacitance density in a given CMOS technology, so an anti-parallel pair of MOSFETs (see Fig. 32) can substitute a linear capacitor with a capacitor that has mediocre linearity but is much smaller. This structure has the additional benefit of symmetric parasitic capacitors at both terminals.

An active inductor can be simulated by transistors using gyrator principles [17], offering a significant area reduction compared to a passive implementation based on a coil constructed from the metal layers. An exemplary implementation based on two MOSFETs is shown in Fig. 33.

VII. Analog Signal Processing

The processing of analog signals is an often needed task when devising circuits. The area-efficient division of voltages (see Fig. 34) or currents (either employing a current mirror as in Fig. 7 or the advanced usage of the Bult current divider,

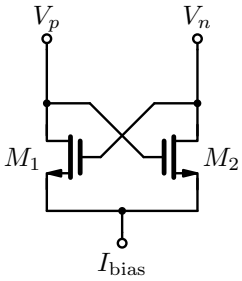

Fig. 28. Two transistors with cross-coupling form a negative resistance between V_p and V_n, mainly employed in oscillators and comparators. As in Fig. 8, the bias current source can be replaced by a fixed potential.

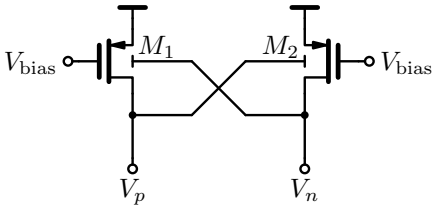

Fig. 29. This circuit is a low-voltage version of Fig. 28, where the body controls the MOSFET, avoiding the significant V_{GS} drop at V_p and V_n [18].

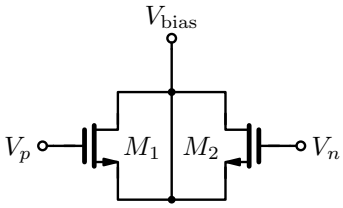

Fig. 30. The varactor (the capacitance between V_p and V_n depends on the bias voltage V_{bias}) is often used in voltage-controlled oscillators. In most technologies, the NMOS can be put inside the *n*-well so that the varactor works in accumulation, providing an optimized tuning range and high Q [19].

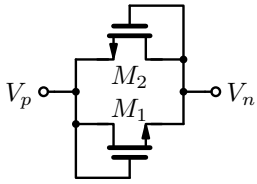

Fig. 31. The anti-parallel MOSFET diodes can be employed for many things, for example, voltage clamping.

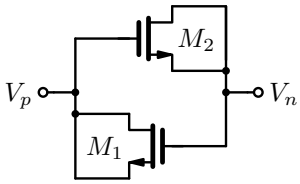

Fig. 32. The anti-parallel MOSFET capacitors make the differential capacitance more linear and symmetrical. As in Fig. 30, an NMOS–in–*n*-well is an option.

shown in Fig. 35) can become handy. In any implementation of an analog-to-digital converter (ADC), a sample-hold stage is required, often exploiting the excellent switching capabil-

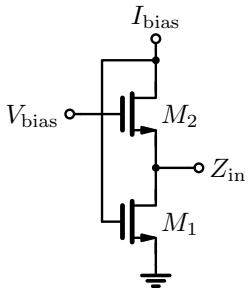

Fig. 33. This circuit, which is a similar configuration as the flipped voltage follower in Fig. 47, can function as an active inductor, providing $L = C_{\text{GS1}}/(g_{m1} \cdot g_{m2})$ [20].

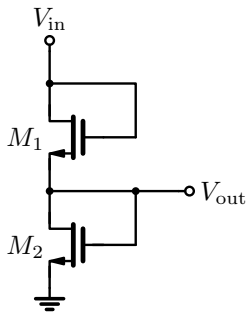

Fig. 34. An area-efficient voltage divider. If M_1 and M_2 are of the same size, then $V_{\text{out}} \approx V_{\text{in}}/2$. Often, a PMOS version is a better choice since it can avoid the body effect by tying the body to the respective source for M_1 and M_2.

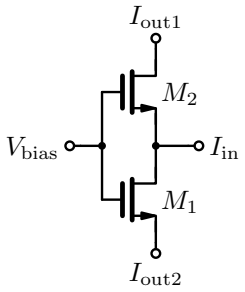

Fig. 35. The Bult current divider (if M_1 and M_2 are of identical size, then I_{in} is precisely split in half between I_{out1} and I_{out2}) [22].

ities of the MOSFET, as demonstrated in Fig. 36, where a MOSFET configured as a capacitor stores the sampled voltage. Sampling can be used for frequency conversion, which can also be achieved by the arrangement in Fig. 37, where the time-variant change of the transconductance of M_1 causes a mixing effect [21].

Realizing the MOSFET operating as a controlled resistor in the triode region, Fig. 36 can be easily transformed into a (programmable) low-pass filter, as shown in Fig. 38. By rewiring the sampling switch into the diode-equivalent MOSFET configuration, a voltage peak detector (Fig. 39) and an approximate voltage doubler (Fig. 40) can be built for ac signals.

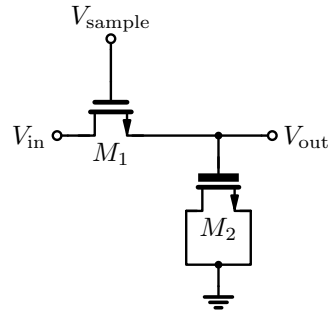

Fig. 36. A sample-and-hold, implementing the gate capacitance of M_2 as a storage capacitor (a low to zero V_{th} would be an advantage in this case) [23].

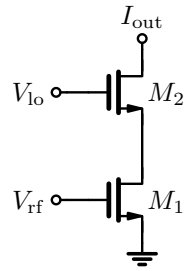

Fig. 37. In the dual-gate MOSFET (a similar arrangement as in Fig. 11), the periodic local oscillator signal V_{lo} causes the time-variant change of the transconductance of M_1, resulting in a frequency conversion from the input V_{rf} to the output I_{out} [21].

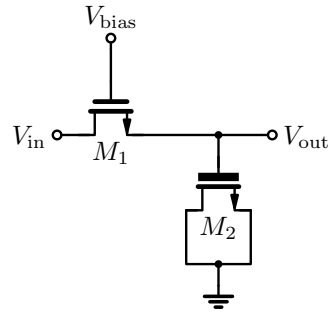

Fig. 38. The circuit of Fig. 36 becomes a continuous-time low-pass filter if M_1 gets a fixed bias instead of a clock signal. Note that this circuit transforms into a high-pass filter when M_1 and M_2 are swapped.

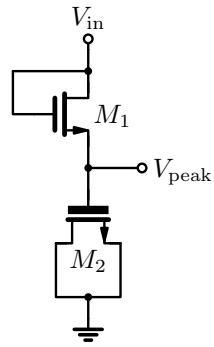

Fig. 39. A voltage peak-detector, where $V_{\text{peak}} = V_{\text{in,max}} - V_{\text{gs1}}$.

VIII. Simple Circuits with a Twist

While most CMOS processes provide thin-film resistors, it is often difficult to realize values in the MΩ to GΩ range. The

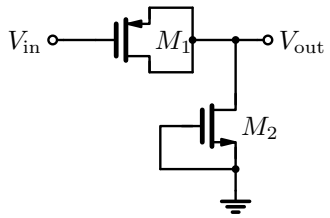

Fig. 40. A similar circuit to Fig. 39 can function as an approximate voltage doubler when driven by a sinusoidal input voltage (on negative swings of $V_{\rm in}$ the capacitor M_1 gets charged to $|V_{\rm in}| - V_{\rm gs2}$, which is added to $V_{\rm in}$ during positive swings when M_2 is off). As in any circuit with negative voltages, proper connection of the wells is required [24].

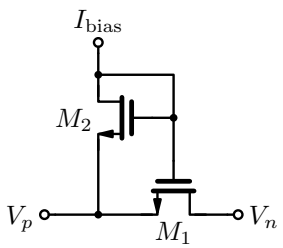

Fig. 41. The controlled (high-impedance) floating resistor.

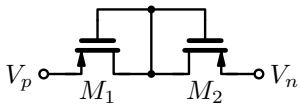

Fig. 42. This arrangement creates an enormous resistance between V_p and V_n, although susceptible to temperature and process variations [25].

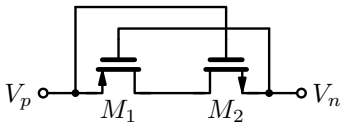

Fig. 43. The ULPD with reduced leakage in the reverse direction when $V_n > V_p$ [26].

MOSFET in triode or off-state (Figs. 41 and 42) is a good alternative at a much smaller silicon area. An improved version of a diode-connected MOSFET is demonstrated in Fig. 43. This two-transistor construction, called the ultra-low-power diode (ULPD), considerably lowers the leakage current in the reverse direction. A level-shift is often needed in a signal path, which can be implemented as pictured in Fig. 44. Sometimes a simple circuit shows a surprising property, and the linear I-to-V converter shown in Fig. 45 is such an example.

Circuits working with the body terminal as a fourth control input can add many additional possibilities, like the logic level-shift implemented in Fig. 46. Feedback is an essential tool in a circuit designer's box: the flipped voltage follower (shown in Fig. 47) and the regulated cascode (see Fig. 48) are significantly improved versions of their simpler counterparts shown in Figs. 9 and 12, respectively.

IX. Using the MOSFET as a BJT

Inherent to the physical structure of a MOSFET is a bipolar junction transistor (BJT), often showing inadequate

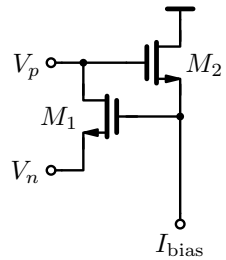

Fig. 44. The floating level shift (or "floating battery") effectively shifts a bias point between V_p and V_n, as $V_{\rm shift} = V_p - V_n = V_{\rm GS1} + V_{\rm GS2}$.

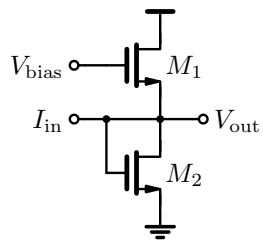

Fig. 45. This circuit is a perfectly linear I-to-V converter with $V_{\rm out}/I_{\rm in} = [\mu C_{\rm ox} \cdot (W/L) \cdot (V_{\rm bias} - 2V_{\rm th})]^{-1}$, if we assume a square-law behavior, and M_1 and M_2 are of same size and kept in saturation (and neglecting body effect) [27].

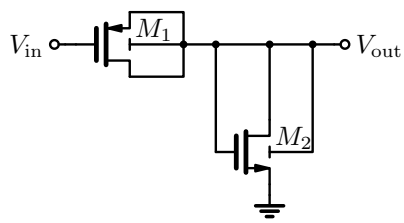

Fig. 46. This circuit shifts a digital input voltage $V_{\rm in}$ to an output voltage $V_{\rm out}$ swinging around $V_{\rm SS}$ [28].

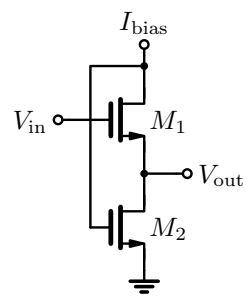

Fig. 47. The flipped voltage follower is an improved version of Fig. 9, employing feedback to lower the output impedance to $r_{\rm out} = g_{ds2}/(g_{m1} \cdot g_{m2})$ [29].

performance like poor current gain $\beta \ll 10$. Still, it can be employed to generate a temperature-dependent voltage, like the circuit shown in Fig. 49, whose proportional-to-complementary temperature (CTAT) property is an excellent addition to the PTAT behavior of the circuit in Fig. 26. Compensating the low β of the parasitic BJT can be achieved by implementing the structure depicted in Fig. 50, which improves the accuracy of a bandgap circuit built from these augmented BJT devices.

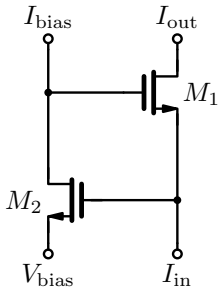

Fig. 48. The regulated cascode, increasing the effect of the cascode M_1 by g_{m2}/g_{ds2} due to feedback. Note that the source of M_2 can be tied to ground if combined with a common-source input stage [30].

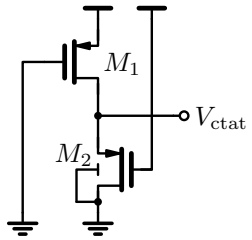

Fig. 49. Exploiting the parasitic (lateral) PNP transistor inherent in a PMOS structure, this simple CTAT voltage generator can be created [31].

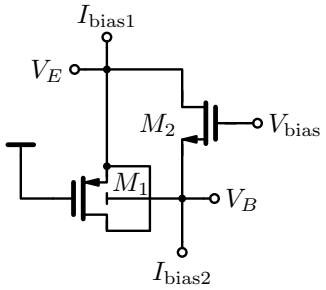

Fig. 50. The parasitic BJT (lateral or vertical), introduced in Fig. 49, often suffers from poor β. This circuit forces the collector current of the parasitic (vertical) PNP to $I_C = I_{\text{bias1}} - I_{\text{bias2}}$, although the collector terminal (being the p-substrate) is not accessible. By doing this, the resulting $V_{\text{EB}} = V_E - V_B$ can be accurately employed in a bandgap circuit [32].

X. Conclusion

Fifty practical circuit snippets, employing just *two* MOS transistors, have been presented. This compilation's motivation is to celebrate the creativity in analog circuit design and demonstrate the versatility of the fabulous MOSFET. Walking through these fifty examples—and considering that this compendium is far from complete [3]—leaves one in awe thinking about the endless possibilities when an analog circuit designer is given a handful of these fantastic transistors!

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